

Bijan Haney

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Summary

Machine learning engineering leader with 10+ years of experience building large-scale data systems and ML infrastructure. Expert in real-time data processing at extreme scale (40M events/sec at CERN, PB+ datasets) and edge-optimized ML deployment for robotics. Proven track record leading teams that deliver end-to-end ML platforms spanning data ingestion, distributed training, and production inference.

Experience

- Robotics Engineer → Engineering Lead → Director of Engineering** Mar 2021 – Present
Viam *New York, NY*
- Lead ML/CV teams building an end-to-end platform for data collection, distributed training, and edge deployment on robots (Raspberry Pi, Jetson Orin Nano/AGX).
 - Scope and prioritize ML infrastructure projects across multiple teams, define technical roadmaps and sprint planning, and mentor engineers.
 - Build real-time inference pipelines connecting video streams to optimized models (Triton, LiteRT, ONNX, TorchScript) with expertise in video compression and model quantization.
 - Employee #10. Built foundational perception libraries (pointcloud processing, reference frames, RealSense integration in C++) and ML modules for edge inference.
 - Deliver production robotics solutions for clients across marine, restaurant, and entertainment industries.
- Senior AI Research Engineer** Apr 2019 – Jan 2021
Augustus Intelligence *New York, NY*
- Trained object detection/segmentation models with PyTorch/Detectron2; R&D on few-shot learning to reduce dataset costs. Deployed CV models for real-time consumer analytics in more than a thousand retail locations across the country on edge devices.
- PhD Researcher – ATLAS Experiment** Jun 2013 – Jan 2019
CERN *Geneva, Switzerland*
- **Large-Scale Data:** Built C++ framework processing terabytes of data (both real and simulated) for Higgs cross section analysis (12% of 30K-line codebase used by a dozen researchers).
 - **Real-Time Streaming:** Filtered 40M collisions/sec to 1K saved events/sec, preserving rare signals. Reduced data by 99.97% while still saving 99.5% of the desired signal for the analysis.
 - **Operations Coordinator:** Led 5-person team for detector operations at 100 kHz; upgraded DAQ hardware for Run 2; implemented CI/CD for codebase; developed and ran calibrations to fix errors due to radiation.

Education

2018, **Ph.D. in Particle Physics**, University of Pennsylvania
2012, **B.S. in Physics (Intensive)**, Yale University

Technical Skills

ML/AI: PyTorch, TensorFlow, LiteRT, ONNX, Triton, claude-code, model quantization/distillation
Data/Infra: PB+ data pipelines, real-time streaming, SQL, distributed systems, CI/CD, Docker, Vertex AI
Robotics: Universal Robots arms, depth cameras/Lidars (RealSense, ORBBEC), RTSP, Raspberry Pi, Jetson, FPGA
Languages: Python, C++, Go

Select Publications & Awards

2025 Breakthrough Prize in Fundamental Physics – For ATLAS LHC Run-2 research (2015–2024)
CVPR 2020 (FGVC7): *Fine-Grain Few-Shot Vision via Domain Knowledge as Hyperspherical Priors*
Phys. Lett. B (2018): Combined Higgs cross section measurements in $H \rightarrow \gamma\gamma$ and $H \rightarrow ZZ^* \rightarrow 4l$ channels